Darshan UNIVERSITY

योग: कर्मसु कौशलम्

Python Programming - 2301CS404

Lab - 14

Roll No.: 352

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Import matplotlib below

In [10]:
```
import matplotlib.pyplot as plt
```

01) WAP to display the line chart for given data for x and y axis.

In [12]:
```
x = range(1,11)
y = [1,5,9,7,5,6,3,2,4,9]
```

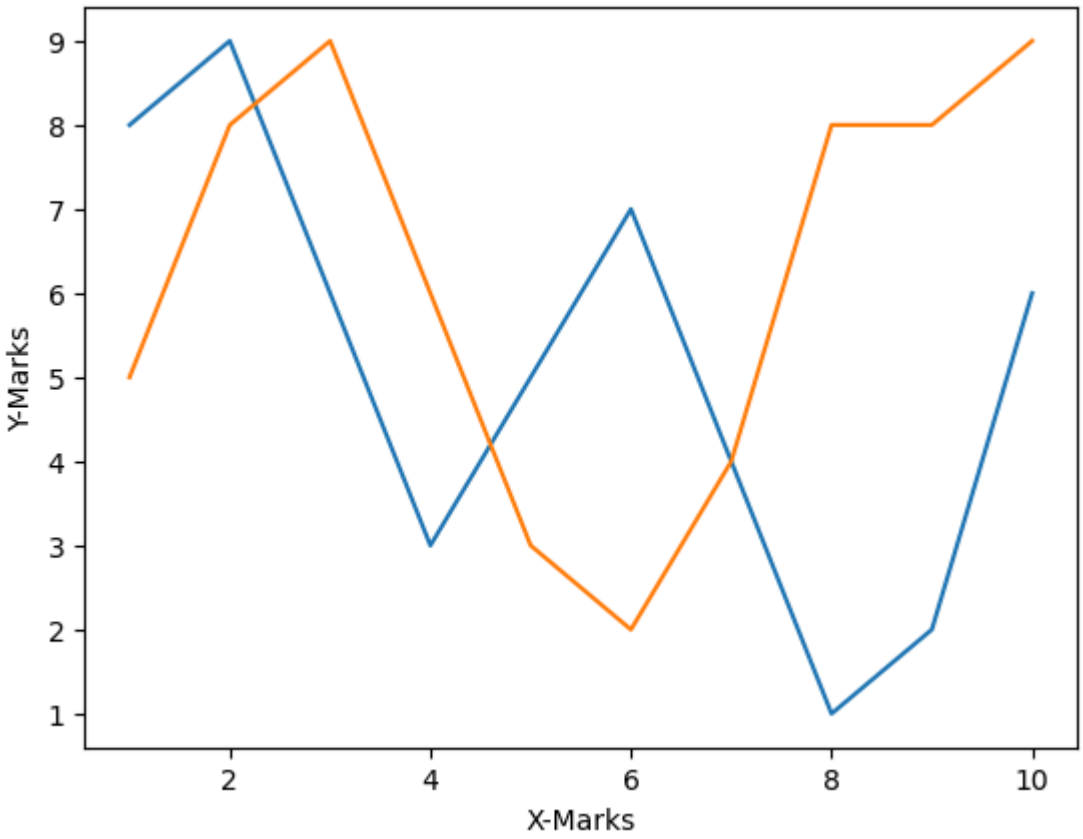
In [13]:
```
# %matplotlib qt <-- show in new tab
plt.plot(x,y)
plt.xlabel("X")
plt.ylabel("Y")
plt.show()
```

In [22]:
```
# %matplotlib inline  <-- show in this cell
```

02) WAP to generate line chart for given data.

In [17]:
```
x = range(1,11,1)
cxMarks= [8,9,6,3,5,7,4,1,2,6]
cyMarks= [5,8,9,6,3,2,4,8,8,9]
```

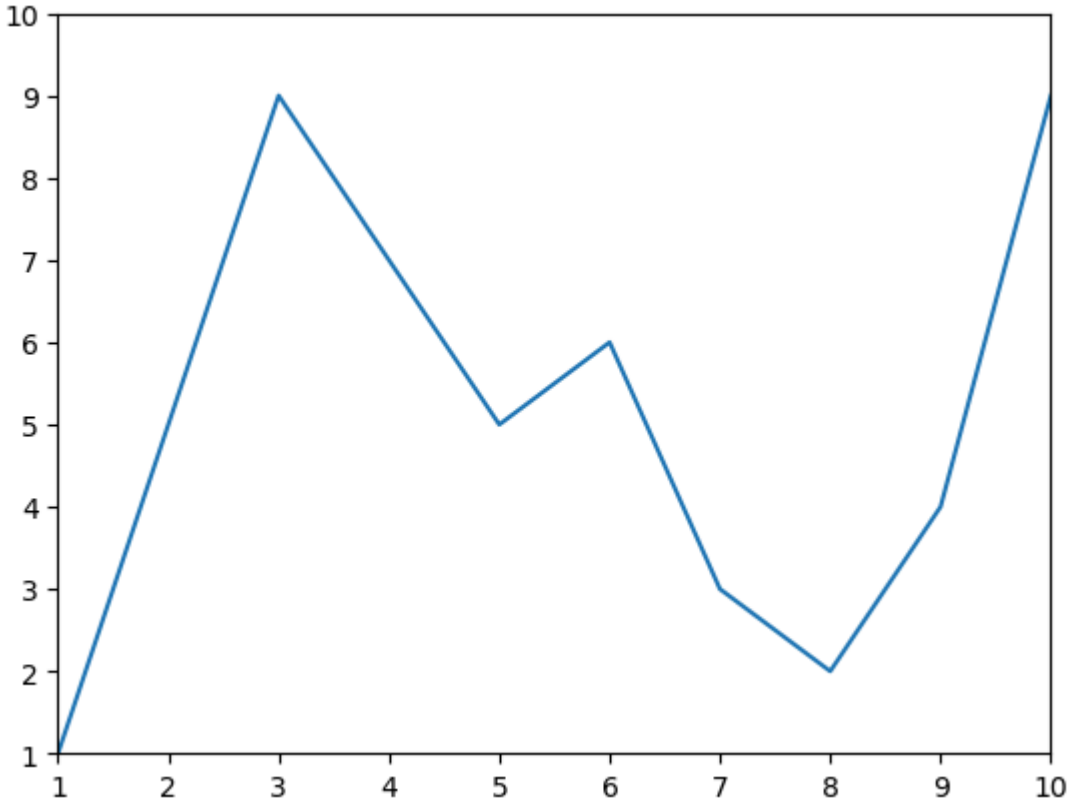
In [23]:
```
plt.plot(x,cxMarks)
plt.plot(x,cyMarks)
plt.xlabel("X-Marks")
plt.ylabel("Y-Marks")
plt.show()
```



03) WAP to generate line chart for given data, but create your custom axis.

In [28]:
```
x = range(1,11)
y = [1,5,9,7,5,6,3,2,4,9]
```

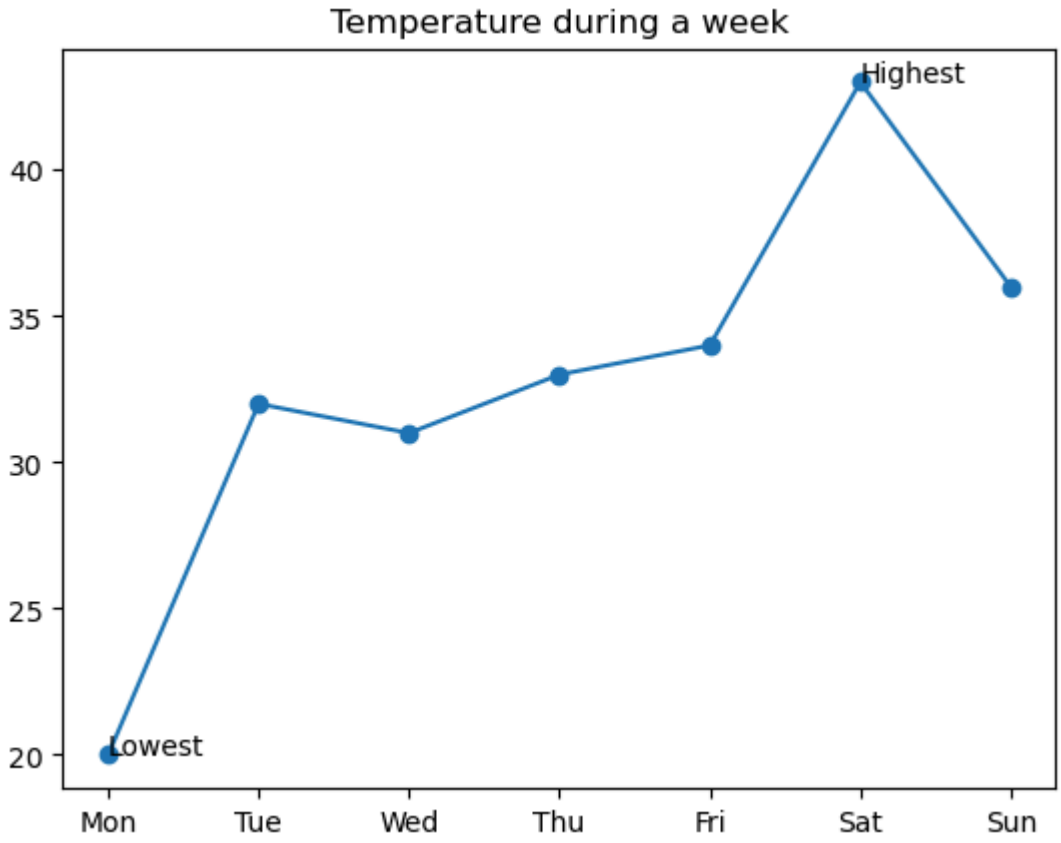
In [33]:
```
ax = plt.axes()
ax.set_xlim([1,10])
ax.set_ylim([1,10])
ax.set_xticks(range(1,11,1))
ax.set_yticks(range(1,11,1))
plt.plot(x,y)
plt.show()
```



04) WAP to plot a line chart showing temperature variation during a week. Mark the lowest and highest temperatue.

In [36]:
```
Days = ["Mon", "Tue", "Wed", "Thu", "Fri", "Sat", "Sun"]
Temperature (°C) = [20, 32, 31, 33, 34, 43, 36]
```

In [49]:
```
plt.plot(Days,Temperature, marker='o')
plt.annotate("Highest", xy = ["Sat",43])
plt.annotate("Lowest", xy = ["Mon",20])
plt.title("Temperature during a week")
plt.show()
```

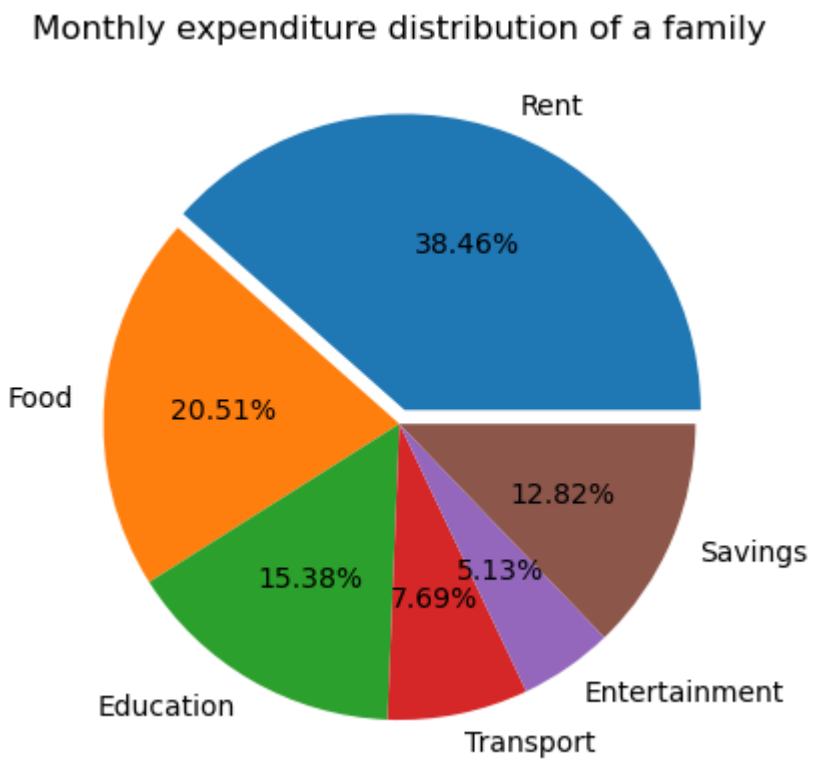


05) WAP to create a detailed pie chart representing the monthly expenditure distribution of a family.

The chart should display labels, percentage values, title, different colors for each slice, and explode one slice to highlight the highest expense.

In [55]:
```
Categories = ["Rent", "Food", "Education", "Transport", "Entertainment", "Saving
Amount (₹) = [15000, 8000, 6000, 3000, 2000, 5000]
```

In [70]:
```
plt.pie(Amount, labels = Categories, autopct = "%.2f%%", explode=[0.05,0,0,0,0,0
plt.title("Monthly expenditure distribution of a family")
plt.show()
```

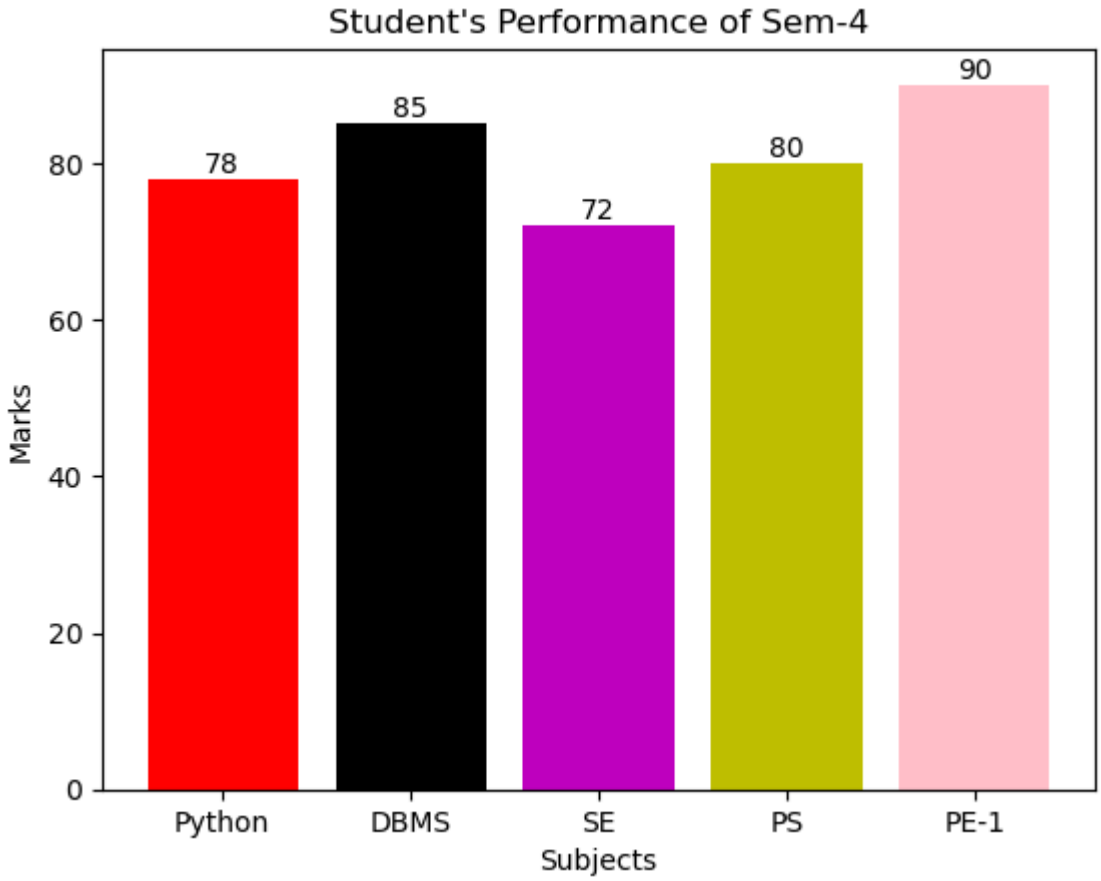


06) WAP to create bar chart showing the marks obtained by students in different subjects.

The program should also display the marks on the top of the respective bar.

In [74]:
```
Subjects = ["Python", "DBMS", "SE", "PS", "PE-1"]
Marks = [78, 85, 72, 80, 90]
```

In [76]:
```
bars = plt.bar(Subjects, Marks, color = ["r","k","m","y","pink"])
plt.bar_label(bars)
plt.xlabel("Subjects")
plt.ylabel("Marks")
plt.title("Student's Performance of Sem-4")
plt.show()
```

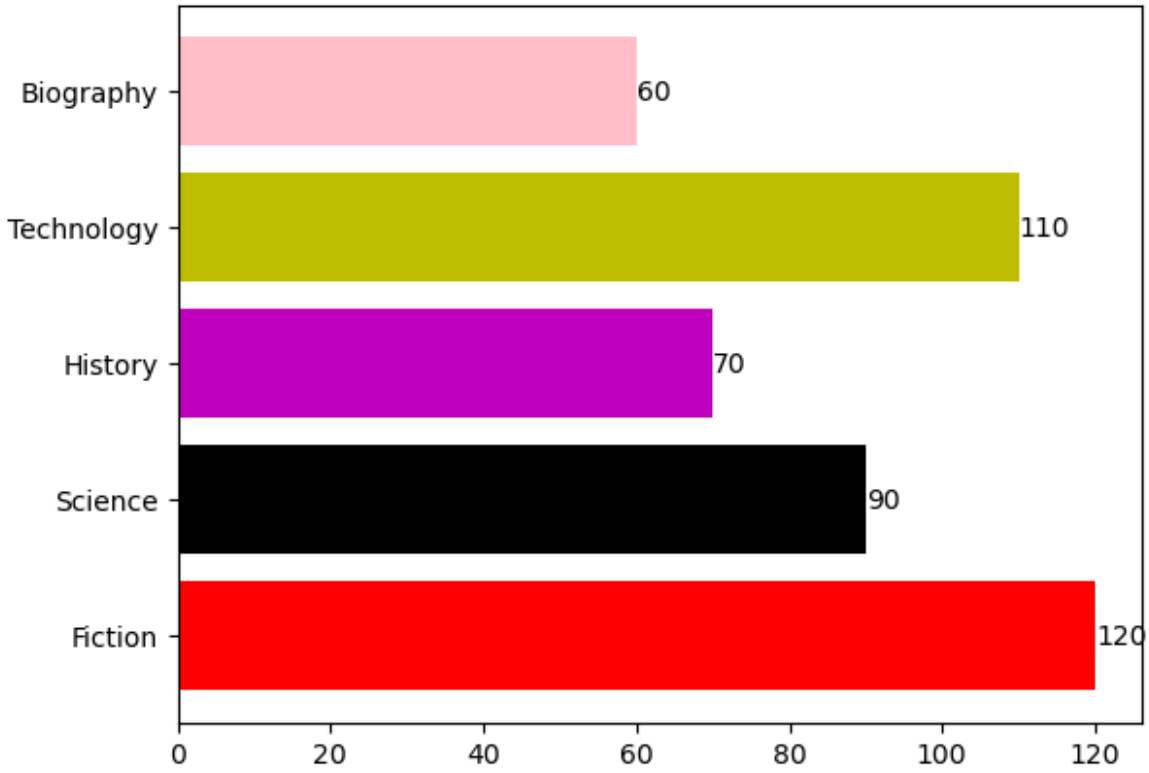


07) WAP to create a horizontal bar chart representing the number of books sold in different categories.

The program should also display the number of books sold at the end of the respective bar.

In [78]:
```
Categories = ["Fiction", "Science", "History", "Technology", "Biography"]
Books Sold = [120, 90, 70, 110, 60]
```

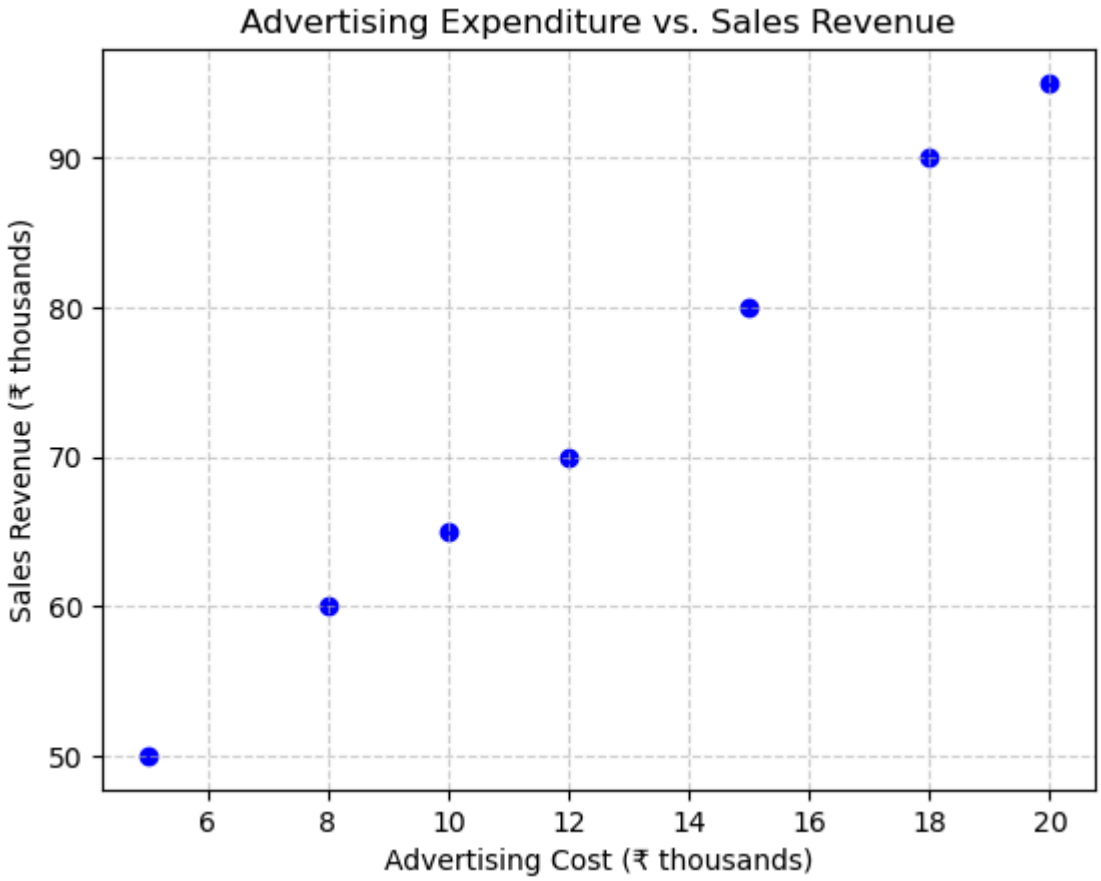
In [79]:
```
bars = plt.barh(Categories, Books, color = ["r","k","m","y","pink"])
plt.bar_label(bars)
plt.show()
```



08) WAP to create a scatter plot representing the relationship between advertising expenditure and sales revenue of a company.

In [82]:
```
Advertising Cost (₹ thousands) = [5, 8, 10, 12, 15, 18, 20]
Sales Revenue (₹ thousands) = [50, 60, 65, 70, 80, 90, 95]
```

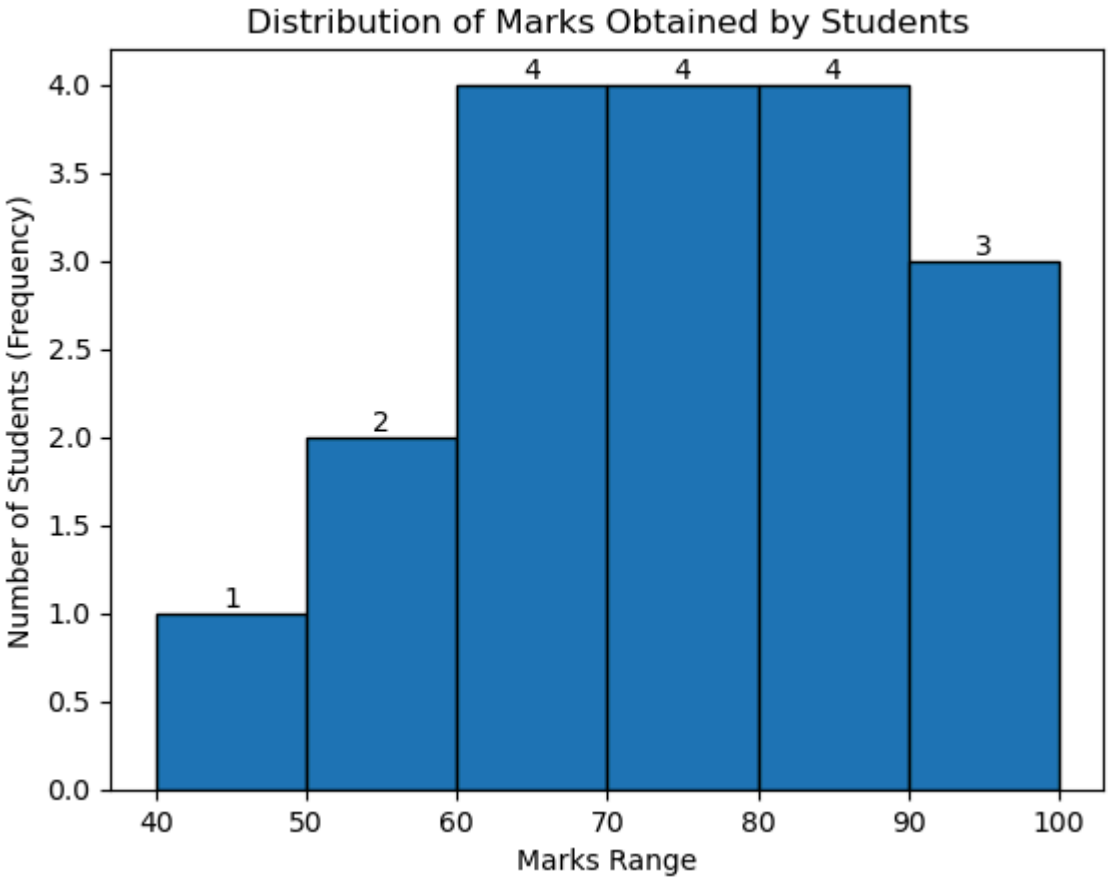
In [104…
```
plt.scatter(Advertising, Sales, color='blue', marker='o')
plt.xlabel("Advertising Cost (₹ thousands)")
plt.ylabel("Sales Revenue (₹ thousands)")
plt.title('Advertising Expenditure vs. Sales Revenue')
plt.grid(True, linestyle='--', alpha=0.6)
plt.show()


# plt.figure(figsize=(8, 5))
# plt.scatter(Advertising, Sales, color='blue', marker='o')

# # Add labels and title
# plt.title('Advertising Expenditure vs. Sales Revenue')
# plt.xlabel('Advertising Cost (₹ thousands)')
# plt.ylabel('Sales Revenue (₹ thousands)')

# # Add grid for better readability
# plt.grid(True, linestyle='--', alpha=0.6)

# # Display the plot
# plt.show()
```



09) WAP to create a histogram representing the distribution of marks obtained by students in a class.

In [85]:
```
Marks = [45, 50, 55, 60, 62, 65, 68, 70, 72, 75, 78, 80, 82, 85, 88, 90, 92, 95]
```

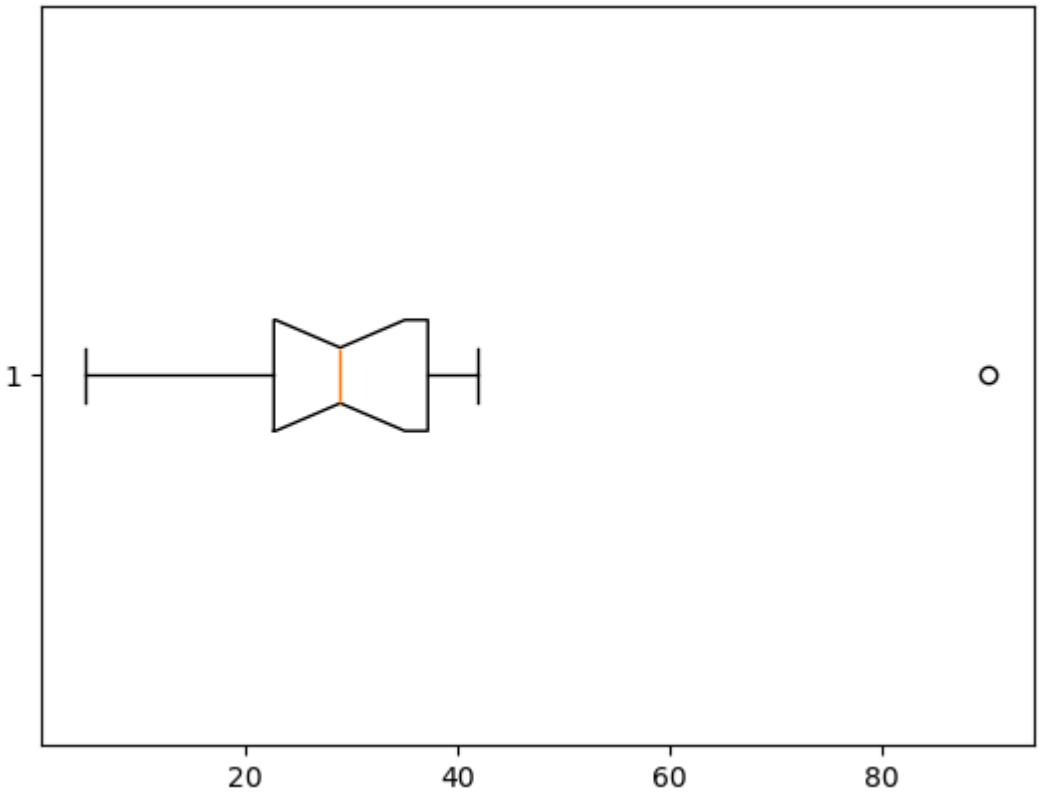
In [107…
```
x,y,bars = plt.hist(Marks,bins=[40, 50, 60, 70, 80, 90, 100], edgecolor='black')
plt.title('Distribution of Marks Obtained by Students')
plt.xlabel('Marks Range')
plt.ylabel('Number of Students (Frequency)')
plt.bar_label(bars)
plt.show()
```



10) WAP to create a box plot representing the distribution of daily commuting time (in minutes) of employees and identify outliers (if any).

In [88]:
```
Commuting Time (minutes) = [5, 18, 20, 22, 25, 27, 28, 30, 32, 35, 38, 40, 42, 9
```

In [96]:
```
plt.boxplot(Commuting, vert = False, notch=True)
plt.show()
```


1
20
40
60
80